

Research Paper

Microcosm fabrication platform for live microscopy of plant-soil systems



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ARTICLE INFO

Keywords:

Microcosm

Plant-soil interactions

Microfabrication

Live imaging

ABSTRACT

Our ability to fully understand how plants acquire water and nutrients from the soil is constrained by the limitations of current technologies. Soil structures and properties are complex, dynamic, and profoundly modified by root and microbial secretions. Detailed descriptions of soil properties are rarely available to the researcher because natural soil is opaque, making direct observations challenging. To address these experimental difficulties, microcosm systems dedicated to live imaging of rhizosphere processes in highly controlled environmental conditions were developed. The system is based on fluorinated granular materials with low refractive indices, termed transparent soils. Microcosm chambers were assembled using poly(dimethyl siloxane) parts (PDMS) fabricated by injection moulding and subsequently joined to glass slides. The control of liquid fluxes in the microcosm was achieved by syringes passing through the PDMS parts or through custom made PDMS sponges. The platform was tested for live imaging experiments using Light Sheet Fluorescence microscopy. Results demonstrated that the platform is suitable for a diverse range of experiments, including live observation of plant roots, split-soil systems and investigations into the effects of soil heterogeneity, controlled water content experiments, and dye tracer monitoring. The technique was used to quantify the increase in infiltration rate due to the presence of roots in soil. This study demonstrates the potential of combining new materials and micro-fabrication techniques to overcome current limitations on plant-soil interaction research.

Nomenclature table

Symbol	Unit	Description
PDMS	NA	Poly(dimethyl siloxane)
FEP	NA	Fluorinated ethylene propylene
h	m	Height of water in the chamber
e	m	Thickness of the porous PDMS layer
ρ	kg m ⁻³	Density of water

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Symbol	Unit	Description
P	Pa	Suction applied to the bottom of the chamber to determine the hydraulic conductivity using the falling head method
K	m s ⁻¹	Hydraulic conductivity
C	ml ml ⁻¹	Concentration of the dye solution
S		Saturation value in HSV image
V		Brightness value in HSV image
k	s ⁻¹	Permeability of the dry transparent soil layer not to the diffusion of the dye

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<https://doi.org/10.1016/j.biosystemseng.2025.02.006>

Received 28 February 2024; Received in revised form 7 February 2025; Accepted 14 February 2025

Available online 8 March 2025

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1. Science4Impact statement

The agricultural sector is looking for ways to develop cropping systems that reduce soil pollution and are more resilient to climate change. This is a major challenge due to the difficulty of measuring processes and compounds in the soil. The techniques presented here miniaturise soil microcosm experiments. They could be used to speed up commercial screens to discover new crop varieties, agrochemicals and inoculants that are better adapted to future environmental conditions.

2. Introduction

Soil microscopy has traditionally been performed on thin soil sections obtained from samples embedded in resin (Tippkötter & Ritz, 1996), chemically processed to identify the location of specific organisms (Nunan et al., 2001). Using sectioning and *in situ* hybridisation techniques, it was possible to reveal the variation in microbial community composition of a Siberian tundra soil with depth (Kobabe et al., 2004) or to localise bacterial species in lichens from soil crusts (Muggia et al., 2013). However, reconstruction of a volume dataset from sectioning techniques is time consuming because of the need to section the sample physically. Also, processes taking place in soil pores are dynamic, and the technique of embedding and sectioning soil samples, though giving important insights into soil structure and composition, is not suitable to fully describe biological mechanisms. Live microscopy techniques can overcome these limitations, but are limited in soil studies due to the opacity of soil, which restricts observations to surface layers (Ritz & Young, 2004). Imaging techniques based on penetrating radiations can overcome these limitations. For example, X-ray CT scanners have notably seen their resolution increase (van Veelen et al., 2020; Zhou et al., 2021), and phase data or high coherence beams help to resolve anatomical features from the low contrast generated by biological material (Kunishima et al., 2020; Sedighi Gilani et al., 2014; Verboven et al., 2015). Unfortunately, X-ray CT does not resolve sub-millimetre biological structure in water (for example soil microorganisms) and X-ray microscopes may not yet permit real-time imaging due to ionising radiation damage made to living tissues.

Since live imaging of natural soils is limited, it has been proposed that artificial soil systems could be developed to reproduce the physical and chemical soil properties whilst simultaneously enabling live observations. One possibility is to use microfabrication techniques to engineer microcosms with a precisely defined arrangement of pores (Stanley et al., 2016). Various pore sizes and geometries can be imprinted on a polymer film that is in turn attached to a glass slide for direct observation. At present the behaviour of diverse microorganisms is commonly microscopically studied in such systems (Mafla-Endara et al., 2021). However, the physical obstacles imprinted are rigid, and the small pores suited to the study of microorganisms are not in turn suitable for studying plant roots, which need to displace soil particles and create macropores that are millimetres in size (Aufrecht et al., 2022). Transparent soil techniques, first proposed by Downie et al. (2012), have overcome these limitations. They consist of granular substrates consisting of materials chosen for their chemical and physical properties. In particular, the materials must have high transparency, and a refractive index that can be matched with an aqueous solution that is not toxic to plants or soil microorganisms. The chemical properties of the surfaces of the particles are also critical and must facilitate repeated rewetting. This has been achieved through coatings with materials characterised by low contact angle (hydrophilic) (Patko et al., 2023) or using materials with ion exchange properties (Downie et al., 2012). Such materials can be mixed with gas and nutrient solutions to create physical conditions close to those of natural soil, leading to similar root physiology compared to plants grown in real soil. Live observations within such systems are possible with visible light due to a refractive index in particles which

matches that of the soil solution (O'Callaghan et al., 2018).

Transparent soil and microfluidic devices are ideally suited to modern light microscopy and together hold great promise for the advancement of rhizosphere science. Light microscopes are widely available instruments in biology laboratories. Modern scientific cameras and detection methods can record extremely weak signals from a range of techniques, e.g. stimulated emission depletion, fluorescence lifetime, Raman spectroscopy, or scattering (Becker, 2012; Freudiger et al., 2008; Hell & Wichmann, 1994; Yang et al., 2013) which increase the richness and resolution of information collected from a sample. Optical sectioning has also been used to obtain data from plant roots and soil microbes *in situ* using transparent soils (Engelhardt et al., 2022; Liu et al., 2021).

However, to fully mimic the microcosms in which plant roots grow, environmental conditions need to be spatially controlled during experiments. Transparent soil microcosms must be compatible with the imaging stage of the microscope, a restriction which can limit the size of the samples being studied and the ability to control growth conditions. Finally, there are limited soil microcosm solutions available “off the shelf” and significant methodological development is needed prior to the running of microcosm experiments.

To address these limitations, this study presents a modular platform for the engineering of microcosm systems suited to live plant-soil microscopies. The study describes designs and protocols for the fabrication of different types of microcosm chambers and test these designs with a case study where the role of plant roots on water infiltration is demonstrated.

3. Material and method

3.1. Injection moulding of PDMS parts of the microcosm

To control fluid flux in and out of the microcosm (of water, nutrients and microorganisms, matching liquids, dyes etc.), microcosm chambers are assembled from glass parts through which microscopic observation and measurements are made, sandwiching poly(dimethyl siloxane) (PDMS) through which needles can be introduced for the injection and aspiration of liquid solutions and suspensions into and out from the transparent soil (Fig. 1). Moulds for the fabrication of PDMS spacers were made from three parts, each of which was fabricated using a CO₂ laser cutter from methacrylate sheets (Stockline Plastics Ltd, UK). The technical drawings used for laser cutting are provided as supplementary information (section 1 and at <https://zenodo.org/deposit/8374546>).

The central part of the mould (red parts, Fig. 2 A) defines the geometry of the PDMS spacer, and the other two lateral parts (blue parts, Fig. 2 A) are used to seal the mould. The thickness of the methacrylate sheet defines the thickness of the transparent soil layer allowed in the microcosm. Thicknesses between 2 mm and 5 mm were tested. 3 mm-thick sheets were found to be a good compromise between the depth of imaging obtainable and the ease of filling of the transparent soil into the chamber system. The moulds were designed for the batch production of chambers. The position and size of the holes used for fastening the moulds were standardised and allowed the alignment of multiple moulds and fastening of the moulds in a single operation using M6 screws. Bespoke spacer geometries could be fabricated (Figs. 1 and 2) to spatially define the transparent soil domain that a root system can explore. When the central part of the mould has a U-shape (Fig. 2 A, centre and right), the domain of transparent soil defined by the spacer is rectangular. To establish distinct compartments and control interactions between them, the spacer fabrication must be conducted in steps. First the enveloping part of the spacer (U shape) is obtained using the mould's central part with a rectangular shape, where a T-shaped plate is introduced to define the width of the PDMS spacer. After the enveloping part of the spacer has cured, the T-shaped plate is substituted by a shorter one to define the position of the next horizontal barrier (Fig. 2 A, left). This system was used to engineer porous barriers within the microcosm

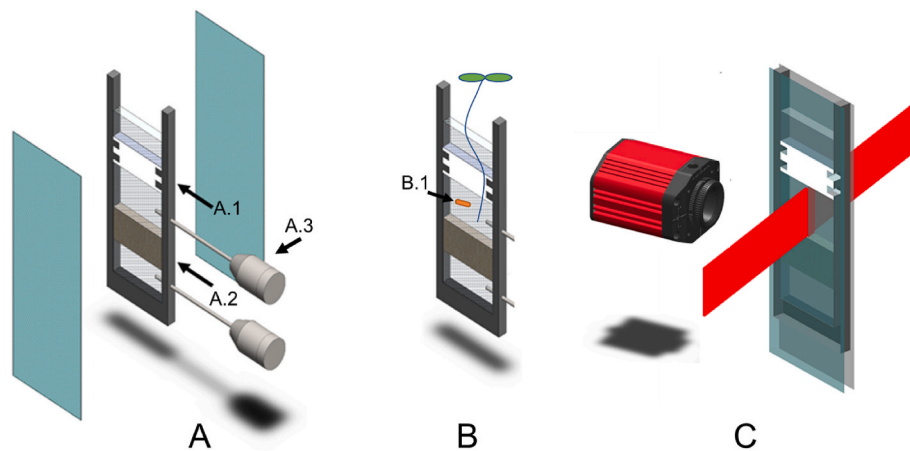


Fig. 1. Microcosm fabrication platform. (A) The chambers consist of two glass slides (blue, left and right) separated by a PDMS spacer (grey). The design of the spacer allows to include layers of soil with different properties (A.1), or semipermeable barriers that regulate interactions between compartments within the microcosm (A.2). The exchange of liquid and microorganisms can also be controlled by infiltration through the PDMS spacer (A.3). (B) Microcosm chambers filled with transparent soils and soil microorganisms (B.1) can then be used to study the roots of growing seedlings. Observation of biological processes is then possible through the glass slides, here using a light sheet microscope (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

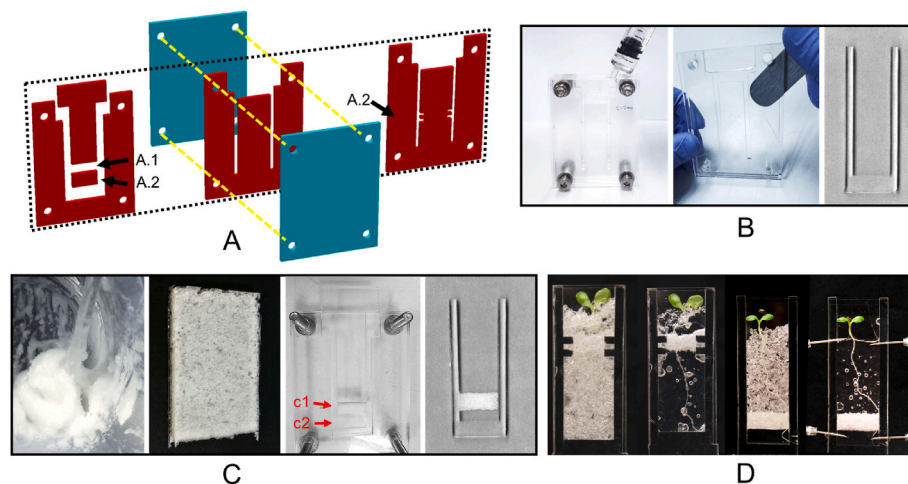


Fig. 2. Design and assembly of the microcosm chambers. (A) PDMS spacers were made by injection moulding, and the mould assembled from 3 parts. The part in the centre (red) defined the geometry of the PDMS spacer. The parts on the sides (blue) were used to close mould. The simplest spacers had a U shape (centre) and defined a parallelepipedic soil domain. To create compartment within the microcosm chamber (left), the part in the centre allowed inserting mobile parts (A.1, A.2) so that barriers separating the compartments can be moulded in a second step. Other spacer geometries were tested to introduce different soil layers (A.3). (B) PDMS injection (left) opening (centre) and extraction from the central part of the mould (right). (C) Porous PDMS assemblages built to impose a tension at the base of the soil following the principle of tension table (Klute & Dirksen, 1986). (D) Different soil properties can be introduced and plants are grown into the system. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

chamber (section 2.2). A detailed protocol for injection moulding of PDMS is provided as supplementary information (section 2.4).

3.2. Fabrication of porous PDMS spacers for fabrication of split chambers

Porous PDMS layers were used to engineer compartments separated by semi-permeable barriers. The barriers separated the soil domain (top) from a water compartments (bottom) where suction can be imposed (bottom) and transparent soil water (top) regulated using the tension table principle (Klute & Dirksen, 1986). Porous PDMS was fabricated from SYLGARDTM 184 silicone elastomer base with initiator mixed in a 10:1 ratio and porogens made of granulated sugar (sucrose) to generate pores (Cha & Kim, 2011; Choi et al., 2011). PDMS:sugar mixtures were tested in three different proportions, 2:1, 1:1 and 1:2 v:v (volume:volume) respectively. Mixtures were then cured in an oven before immersion in water to dissolve the porogen. The size of pores in PDMS was

controlled using milled and sieved sugar granules with three size categories: small (0.25 mm–0.5 mm), medium (0.5 mm–0.71 mm) and large (0.71 mm–1.0 mm). A detailed protocol for the fabrication of porous PDMS material is supplied as supplementary information (section 3).

3.3. Assembly of chambers

Microcosm chambers were obtained by assembling the different parts described in 2.1 and 2.2. Glass-PDMS bonds were achieved by oxygen plasma treatment of PDMS parts, which introduced polar silanol groups (Si–OH) on the PDMS surface in turn producing Si–O–Si bonds at the expense of water molecules (Bhattacharya et al., 2005). Plasma treatment was performed using a workbench plasma system (HPT-100, Henniker, UK) operated for 15 s at 100 W. Following surface treatment, glass and PDMS surfaces were joined together and pressure applied to ensure a strong bond between all adjoining surfaces. The Si–O–Si bonds

are not only robust mechanically, they also provide water tight seals that allows control of water fluxes in the microcosm chambers supplementary information (section 5). The 3D model was designed such that the PDMS spacer would fit onto standard glass microscopy slides (76 mm × 26 mm × 1 mm, VWR, UK). In order to facilitate gas exchange and for oxygen to enter the chamber, it is possible to replace either or both of the glass slides with a PDMS membrane. A detailed protocol for the assembly of glass and PDMS parts is provided as supplementary information (section 2.4).

3.4. Water retention of porous PDMS with the tension table method

To test the ability of porous PDMS to regulate water content in the transparent soil, different hydraulic properties were characterised. First, the water retention of porous PDMS was characterised using the tension table method (EcoTech, Bonn, Germany). Porous PDMS parts were prepared with small, medium and large pore spaces using granulated sugars as porogen. Following immersion in water, porous PDMS parts were dried at room temperature before treatment by oxygen plasma for 15 s (at 100 W). Porous PDMS samples were then saturated with water before placing them in the tension table (Klute & Dirksen, 1986). Samples were weighed daily, and water tension was adjusted when no loss of water was recorded between two measurements. Tensions of 1 kPa, 2 kPa, 5 kPa and 10 kPa were applied and the gravimetric moisture was measured at each pressure point for each piece. Larger suctions were not applied because at 10 kPa, a large part of the water had left the sample. When applying such suction in the water compartment of the microcosm chamber, air started to appear and control of the water matric potential was lost. All experiments were conducted on three different samples for each pore size, and measurements were repeated three times on each sample.

3.5. Hydraulic conductivity of porous PDMS using the falling-head method

The saturated hydraulic conductivity of porous PDMS was characterised using the falling-head method. The chambers were compartmentalised with a 1 cm porous PDMS layer (Fig. 2 D), with the compartment separated with small pores, medium pores and large pores in PDMS, respectively. The chambers were filled with 3.5 cm of water and a tension of constant pressure P was applied and the height of water in the chamber was recorded every 2 min. Detailed characterisation of the porosity of the PDMS showed the structure contains numerous pores with randomly distributed sizes (supplementary information section 6). Therefore, the flow of water through the porous PDMS was represented via the hydraulic conductivity parameter K (m s^{-1}), which is related to the change in the height of water in the chamber h through Darcy's Law,

$$\frac{dh}{dt} = -\frac{K}{e} \left(h - \frac{P}{\rho g} \right) \quad (1)$$

where e (m) is the thickness of the porous PDMS layer, ρ (kg m^{-3}) is the density of water and g (m s^{-2}) is the acceleration due to gravity. Here, the thickness of the porous PDMS layer e (m) was measured by a ruler with an uncertainty of approximate ± 1 mm. The solution to Eq. (1) is of the form

$$h(t) = \left(h_0 - \frac{P}{\rho g} \right) \exp\left(-\frac{K}{e} t\right) + \frac{P}{\rho g}, h > 0 \quad (2)$$

The hydraulic conductivity can therefore be estimated as (Fig. 3 A) from the change with time of the height of water in the chamber h ,

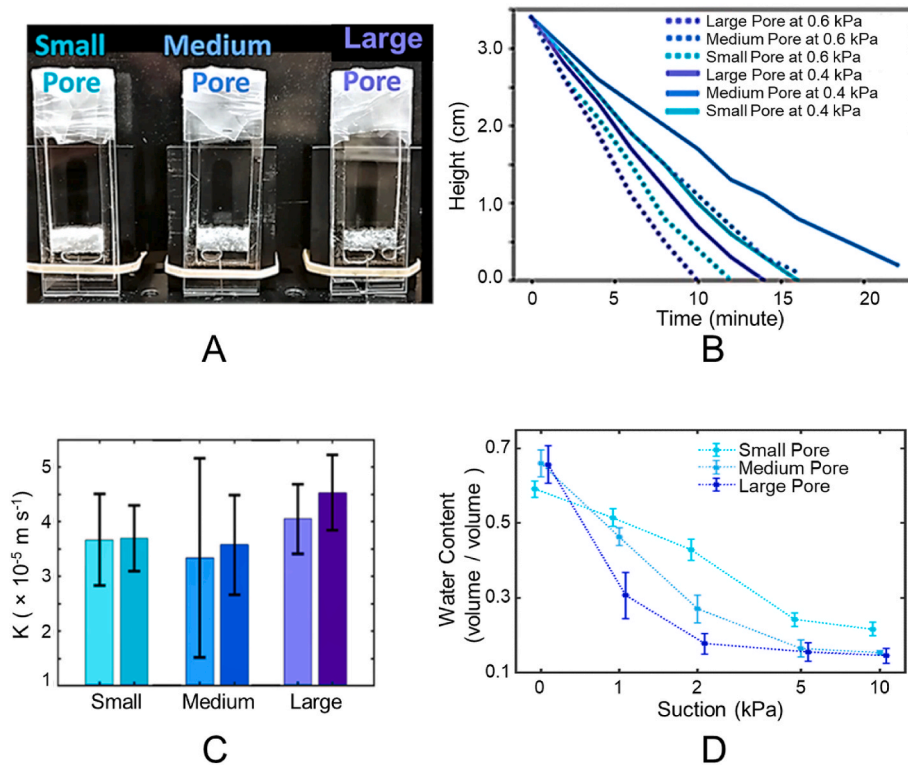


Fig. 3. Hydraulic properties of porous poly(dimethyl siloxane) (PDMS) in microcosm chambers. (A) Set up for falling-head measurement of the hydraulic conductivity K of the porous PDMS. A syringe is used to apply a known tension in the lower container and the height of the water column is measured at regular time interval. (B) Results show that the flux of water through porous PDMS occurs faster with small pores than with medium pores, but slower than with large pores. (C) Resulting hydraulic conductivity. (D) Water retention of the different porous PDMS sponges. Error bars indicate mean $\pm$ standard deviation.

$$K = -\frac{e}{t} \ln \left(\frac{h(t) - \frac{p}{\rho g}}{h_0 - \frac{p}{\rho g}} \right) \quad (3)$$

All experiments were repeated 6 times on the same sample.

3.6. Low refractive index materials and their suitability for growing, and live imaging of, roots

Before plants can be grown in microcosm chambers, water and various types of transparent soil substrates were introduced. The materials tested for the fabrication of transparent soils consisted of Nafion™ particles (IonPower, USA), fluorinated ethylene propylene (FEP) particles (Holscot, UK), hydrogel beads (Ma et al., 2019), and Forblue™ particles (tube particles of two types of: outside diameter: 3 and 0.5 mm, sunsep™ tubes, AGC, Japan). Nafion™ particles (0.25–1.25 mm) were prepared according to previous work (Downie et al., 2012). FEP pellets (approximately 2 mm) were sieved from raw materials. To assess the effect of substrates on growth, a Scottish agricultural soil (sandy loam, 71% sand, 19% silt and 10% clay, Bullion field at the James Hutton Institute, Dundee, 56°27'34" N 3°4'21" W) (Loades et al., 2013) and 1.5% (weight/volume) agar medium (Sigma A1296-100G) were also used. A bio-compatibility assay was implemented to test the suitability of these various materials. Seeds of lettuce (*Lactuca sativa* “all year round”, Sutton Seeds, UK) were surface sterilised by washing in 10% bleach for 15 min followed by thorough rinsing with sterile dH₂O. Sterilised seeds were placed in agar on a Petri dish and incubated at 21 °C in a growth facility for germination. 7 mL of the different processed substrates were poured in glass tubes (16 mm × 100 mm, OD × length) and saturated in water, and seedlings with similar size were grown in 6 replicate experiments.

A second experiment was performed to assess the image quality obtained when using the above-mentioned substrates. The microcosm chambers were first autoclave sterilised. Seedlings were germinated as described in the paragraph above and transferred under sterile laminar air flow. Seedlings were grown for 2 days at 21 °C with 16 h light and 8 h dark period (Fig. 2 D). A colloidal suspension (Percoll®, GE Healthcare, UK) at 80% v/v in dH₂O was used for refractive index matching of sulforrhodamine-B (Sigma-Aldrich, UK) stained at 1 µg/100 mL concentration. When imaging through Nafion™ and Forblue™ particles, light scattering was acquired using 633 nm illumination and recorded without a filter. The stained Nafion™ and Forblue™ particles were illuminated at 561 nm and recorded with a long pass filter (cut-on wavelength: 570 nm, Thorlabs).

3.7. Light sheet fluorescence microscopy

Images were acquired using a custom-made Light Sheet Fluorescence Microscope (LSFM) developed by Liu et al. (2021). A four wavelength-laser source (488 nm, 514 nm, 561 nm and 633 nm, Vortran Versalase, Laser 2000 Ltd, UK) was used to create a thin sheet of light to illuminate the sample. Two counterpropagating light sheets sectioned the sample, one from each of two sides of the chamber. A microscope objective (5×, Mitutoyo, Japan) and a digital camera (Orca flash 4, Hamamatsu, Japan) were placed orthogonally to the light sheet to capture light emitted from the sample, either through scattering or fluorescence, at a maximum dynamic range (16 bits of greyscale). To isolate the fluorescence signal, the imaging arm was respectively fitted with long pass filters of 530 nm, 570 nm, 645 nm and 665 nm (FGL530, FGL570, FGL645, FGL665, Thorlabs, UK).

3.8. Case study: effect of root growth on water infiltration

Plants are known to promote water infiltration, with roots believed to create preferential flow through soil (Jačka et al., 2021; Leung et al.,

2018). However, the mechanisms through which infiltration is facilitated are not known. To investigate how plant roots may facilitate water infiltration, the study focused on how plants affect the vertical movement of tracer dyes through dry transparent soil layers. Chambers were created using 4 mm thick PDMS spacers. Wet transparent soil was added at the bottom of the microcosm chamber. An 8 mm layer of dry transparent soil was introduced, and another wet transparent soil layer was added at the top. The dry layer consisted of Nafion™ particles that were oven dried at 100 °C overnight. Drying of Nafion™ causes the water contact angle to change from 25° (hydrophilic) to 105° (hydrophobic) (Goswami et al., 2008). This behaviour is strongly hysteretic such that rewetting takes several days, during which the central layer remains largely hydrophobic. Winter wheat seedlings (*Triticum aestivum* var. Filon) were sown on the top transparent soil layer. 7 days after sowing, a tracer dye was placed in the upper wet layer. In the control samples only the tracer dye was added, but preparation was otherwise the same. The tracer dye is composed of a concentration of 0.1 mL of red dye (Red food colorant, Vahiné, Spain) in a volume of 1 mL of sterile water. To accurately evaluate the concentration of the dye, an imaging analysis method was developed. Chambers filled with tracer dye solutions concentration of (0, 0.025, 0.05, 1, 2, 4, 6, 8, 10, 12, 14, 16, 18 or 20) × 10⁻⁴ (vol/vol) were imaged by a colour camera (EOS 100, Cannon) in front of a backlight consisting of a white monitor screen. The images acquired by the camera were cropped and transformed to an HSV coordinate system (Hue, Saturation, Brightness or Value). A piecewise function relating the Saturation (S) and Brightness (V) to the dye concentration was then established,

$$\begin{cases} C = f(V); & \text{if } H \in R \\ C = g(S) & \text{otherwise} \end{cases} \quad (4)$$

Where C (ml ml⁻¹) is the concentration of the dye solution and R is the hue range of the calibrated red colour. The algorithm was implemented in MATLAB (MathWorks Inc, USA) and subsequently used to estimate the concentration of the dye in the chamber.

To quantify the infiltration of the tracer dye through the microcosm, a quantitative measurement of the saturation of images was taken daily. Images were acquired with a Canon EOS 550D camera (22.3 × 14.9 mm CMOS, 18.0 megapixels) with a EF-s 18–55 mm lens ($f = 55$ mm, $f/5.6$), at an exposure time of 1/13 of a second. The resulting resolution is approximately (0.02–0.03) mm per pixel. An RGB (Red-Green-Blue) image was obtained and the permeability k (s⁻¹) was calculated using the following formula:

$$k = \frac{(C_2 - C_1) / (C_0 - SC_1)}{dt} \quad (5)$$

The concentration in the upper (C_0) and lower (C_1) soil layers were measured during the first day and the lower region (C_2) was imaged during the following days. dt is the time interval between the measurements of C_1 and C_2 .

4. Results and discussion

4.1. Microfabrication techniques allow to engineer growth microcosms for visualising plant and soil microorganisms

Using the fabrication techniques outlined in section 2.1, 2.2 and 2.3, various chamber designs were tested and constructed (Fig. 2 D). The microcosm chambers provided access for the preparation of custom microcosms for confocal imaging, light-sheet imaging and for optical manipulation of plant roots. This was aided by the maintenance of a constant distance between the two glass parts as set by the U-shapes produced by PDMS spacers. The PDMS spacer allowed easy perforation by needles while maintaining a watertight seal during experiments (Fig. S1). The covalent bonds (Si–O–Si) between PDMS and glass were durable and maintained after several autoclave treatments such that the

chambers were reusable after cleaning. Importantly, the design allowed for the control of fluids inside the microcosms created (Supplementary Figure S1), including through the porous PDMS compartment.

Similar materials and fabrication techniques have been used in other studies. For instance, PDMS has typically been used to engineer structures at a smaller scale (Lockery et al., 2008; Ozoe et al., 2015, pp. 702–705; Patabadige et al., 2019) or cantilever systems (measurement of forces), when forming a nanoporous interface (exudate sampling), and to introduce pores. These systems are less easily transferred to other labs because of the need for specialised equipment and facilities as well as the knowhow to operate them. Simpler fabrication techniques such as that proposed by EcoFAB have been demonstrated (Gao et al., 2018; Sasse et al., 2019). With this framework the study shows that such fabrication techniques can be expanded to combine features that allow for the control of water in the microcosms created (fluxes, content and heterogeneity), which is critical in the study of plant responses to their environment. Fabrication of such chambers could be further facilitated using 3D printing, which can now rapidly build PDMS moulds from centimetres to micrometres in size (Gao et al., 2018; Ge et al., 2021; Liu et al., 2021; Sasse et al., 2019).

4.2. Engineered porous PDMS barriers enable control of water flow and soil water content in the chamber

Various porogen compositions were tested to create tuneable porous PDMS barriers in the system (Fig. 2 D). PDMS with low porogen content (1:2, v:v) did not show permeability to air (it was impossible to blow air through it). This was due to lack of connectivity between sugar particles, which simultaneously prevented the removal of sugar from the inner parts (rigid particles still felt under the PDMS outer surface) whilst not creating the interconnected pore spaces needed for fluid transport. Improvements were achieved with PDMS mixtures with increased sugar content (1:1, v:v), but a complete removal of sugar was also not possible. PDMS with a high sugar content (2:1, v:v) enabled a complete removal of sugar and permeability of the porous material through interconnected pore spaces.

The hydraulic conductivity of the porous PDMS compartment was tested using the falling-head method (Fig. 3 B and C). The results indicated the hydraulic conductivity of PDMS under a tension of 0.4 kPa or 0.6 kPa was $(3.7 \pm 0.8) \times 10^{-5} \text{ m s}^{-1}$ or $(3.7 \pm 0.6) \times 10^{-5} \text{ m s}^{-1}$ with small pores (>0.25 but <0.50 mm), $(3.3 \pm 1.8) \times 10^{-5} \text{ m s}^{-1}$ or $(3.6 \pm 0.9) \times 10^{-5} \text{ m s}^{-1}$ with medium pores (>0.50 but <0.71 mm), and $(4.1 \pm 0.6) \times 10^{-6} \text{ m s}^{-1}$ or $(4.5 \pm 0.7) \times 10^{-5} \text{ m s}^{-1}$ with large pores (>0.71 mm but <1.00 mm), respectively. The flux of water through PDMS occurs faster with small pores than with medium pores but slower than with large pores.

The water retention of porous PDMS with the small pores was greatest compared with the medium pores and large pores, particularly when the tension ranged from 0 kPa to 5 kPa. When the tension reached 10 kPa, the water in the porous PDMS reduced significantly (Fig. 3 D). Application of a tension of 0.4 kPa removed all water from the top part of the chamber. The curves showing the flux of water through porous PDMS showed that pore size consistently affected the transport of water.

The choice of PDMS as material for fabricating the porous separation between water and soil had both advantages and inconvenient. PDMS is an inert material which is thermostable, resistance to oxidation, nontoxic, high solvent compatibility, and compatible with cell cultures. For these reasons, it is found in many microfluidic applications in the life sciences (McDonald & Whitesides, 2002; Ren et al., 2013). We did not observe significant dimensional variability during experiments. This is consistent with previous study which found the coefficient of moisture expansion of porous sponges is almost constant and independent of the porosity and extremely low (3.45×10^{-5}) (Pan et al., 2020). PDMS however is hydrophobic. Even though plasma treatment was applied to reduce the contact angle of surfaces, the treatment typically results in contact angles of 50° (Fritz & Owen, 1995). Developing improved

systems suitable to absorb water more strongly may involve new materials or subsequent modifications of PDMS surfaces.

Overall, the results of the hydraulic characterisation of porous PDMS layers were as expected. Smaller pore sizes increased water retention and reduced the permeability of the porous PDMS. The hydraulic conductivity of PDMS was moderate and similar to that of sandy loam soils (Allen et al., 2010). The water retention, enabled by the porous PDMS, was similar to sand (Durner, 2023). This system may not be suitable to control water content in a natural soil (only wet conditions), but is adequate when using artificial soils such as those made of FEP (Patko et al., 2023) or Nafion™ (Downie et al., 2012), which have similar water retention properties to sand.

4.3. Suitability of materials for the fabrication of transparent soils

Light sheet image data were acquired from a range of substrate types (Fig. 4). Nafion™-based transparent soil stained with sulforhodamine-B produced images where both root and particles could be visualised with great clarity (Fig. 4). Images of similar quality were obtained with Forblue™ based transparent soil. Images acquired from roots grown in FEP based transparent soil showed the lowest quality of those investigated here. FEP is less transparent than other materials tested and image noise from the substrate is most likely due to scattering from within the FEP itself, whilst also coming from Fresnel reflections at air gaps between the FEP and water due to the hydrophobic behaviour of surfaces. Soils made of hydrogel beads were transparent and image quality obtained with the system showed similar properties to those obtained in Nafion™- and Forblue™-based transparent soil.

Overall, plants had a higher dry weight in soil and in Nafion™ than with other artificial substrates (Fig. 4). The primary root length was longer in Nafion™ particles and Forblue™ particles. However, more secondary roots were seen in agricultural soil. Roots grown in Forblue™ particles with 0.5 mm diameter were frequently damaged during extraction. Plants grown in soil, Nafion™, large Forblue™ particles, and hydrogel beads had slightly higher dry weight than those grown in agar. More secondary roots were observed in plants grown in soil, Nafion™, FEP and large Forblue™ particles.

Overall, the results indicated that plant growth in artificial soils more closely resembled growth in natural soil compared to growth in a gel medium, which confirmed previous experiments (Downie et al., 2012; Ma et al., 2019). No plants showed signs of stress and overall, all materials could be considered as biocompatible. Most materials showed capability to provide high quality live images of the rhizosphere. Use of low cost, low refractive index polymers such as FEP produced lower quality data, but the recent methods for functionalising FEP (Patko et al., 2023) indicated potential for significant improvement of the technique.

The results in this study confirm it is possible to construct centimetre-scale microcosms for plants with the ability to measure root development up to one week. The physical model constructed from transparent and soil-like substrates is able to mimic the heterogeneous nature of soil, something which is key to forming physiochemical gradients, and heterogeneity in root exudates and microbial communities (Kuzakov & Razavi, 2019). Indeed, root growth in transparent soils were shown to resemble more closely that of natural soil (Downie et al., 2012), to promote more diverse forms of mobility (Engelhardt et al., 2022; Liu et al., 2021), and produce microhabitats due to the heterogeneous distribution of root exudates (Patko et al., 2023). Spatial and temporal dynamics in transparent soil present a smaller but more dynamic rhizosphere when compared to a gel-based system, where homogenous diffusion might lead to overestimated rhizosphere ranges (Yee et al., 2021) and other non-physiologically representative behaviours.

4.4. Image analysis demonstrates how root growth affects the permeability of dry transparent soil layers

It was possible to create a heterogeneous water distribution by

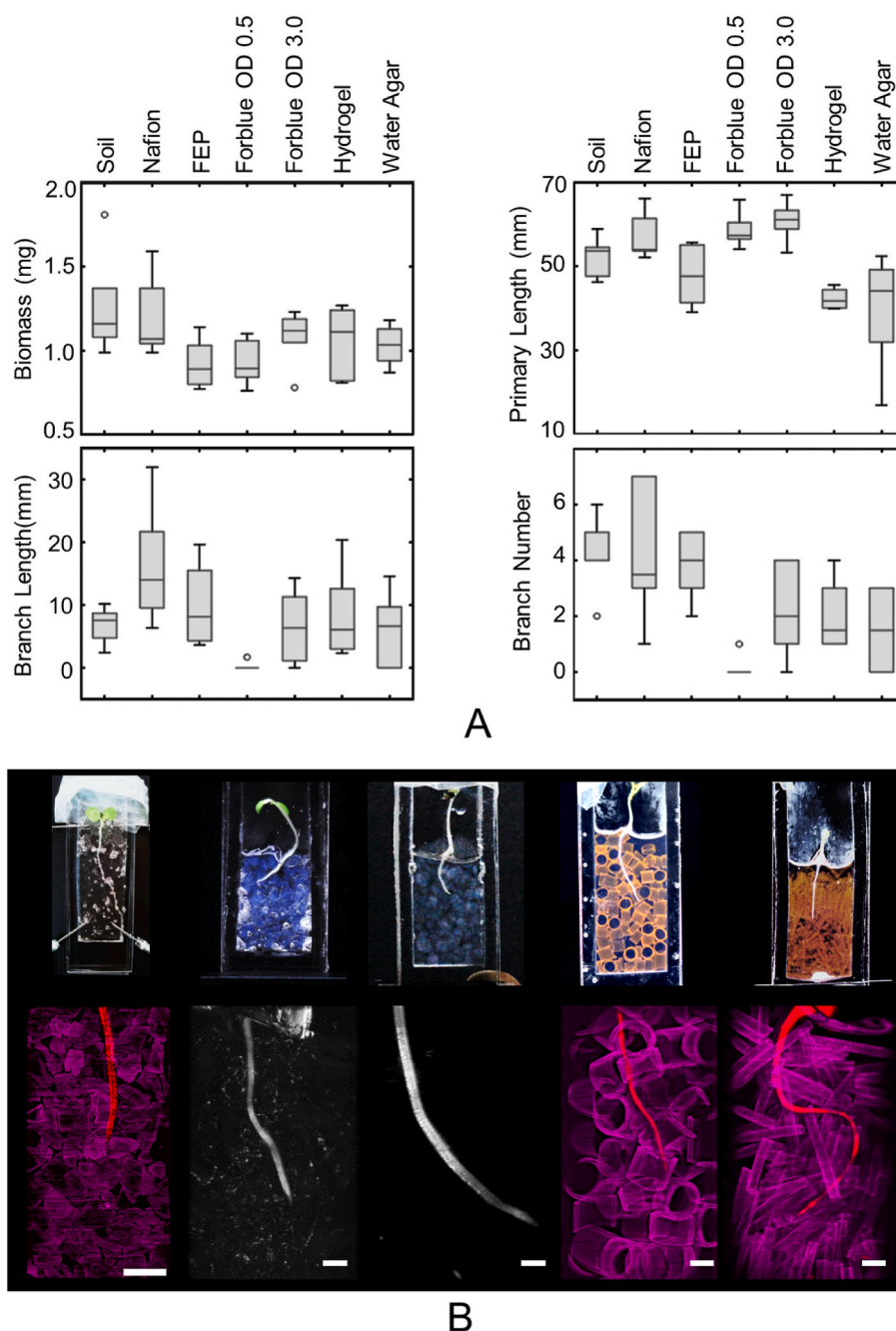


Fig. 4. Testing of different materials for growth and live imaging through artificial soils. (A) Box plot showing the effect of substrate on the growth of lettuce seedlings. (B) Live imaging of lettuce seedling grown in transparent soils made of five low refractive index materials. Top left to right: lettuce seedlings in NafionTM particles, fluorinated ethylene propylene (FEP) particles, hydrogel particles, sulforrhodamine-B stained ForblueTM particles (3 mm outside diameter), and sulforrhodamine-B stained ForblueTM particles (0.5 mm outside diameter). Bottom left to right: maximum projection images of lettuce roots (red) in sulforrhodamine-B stained NafionTM (magenta), lettuce root (grey) in FEP particles (grey), lettuce root (grey) in hydrogel particles (grey but barely imaged), lettuce root (red) in sulforrhodamine-B stained ForblueTM particles of 3 mm OD (magenta), lettuce root (red) in sulforrhodamine-B stained ForblueTM particles of 0.5 mm OD (magenta). All scale bars represent 2 mm. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

adding a hydrophobic transparent soil layer to an imaging chamber (dry NafionTM particle layer as thin as 3 mm in height). The glass window of the microcosm chamber allowed measurements based on quantitative imaging methods developed previously. Here, measurements were made in the HSV colour space and allowed estimation of the permeability of dry transparent soil layers to a dye tracer (Fig. 5A and B). The methods showed there was a nearly linear relationship between dye concentration and either the brightness (in the range [0.00, 0.06]) and saturation (in the range [0.06, 0.20]). The method was applied to quantify the infiltration of water through dry transparent soil with and without

plants (Fig. 5 C). Results showed the permeability of the dry transparent soil layer was reduced when samples contained a growing seedling by comparison to control samples (no seedlings), even though water uptake reduced the pressure head in the top transparent soil layer (Fig. 5 D). 24 h after introduction of the tracer dye, colouration in the bottom transparent soil layer was observed in samples containing roots and the mean permeability of the dry transparent soil layer was $6.0 \times 10^{-5} \text{ mg ml}^{-1} \text{ h}^{-1}$. In the following 72 h, the permeability increased slightly, probably due to the slow rewetting of the dry transparent soil but remained similar to an order of magnitude. A Kruskal-Wallis rank sum test was

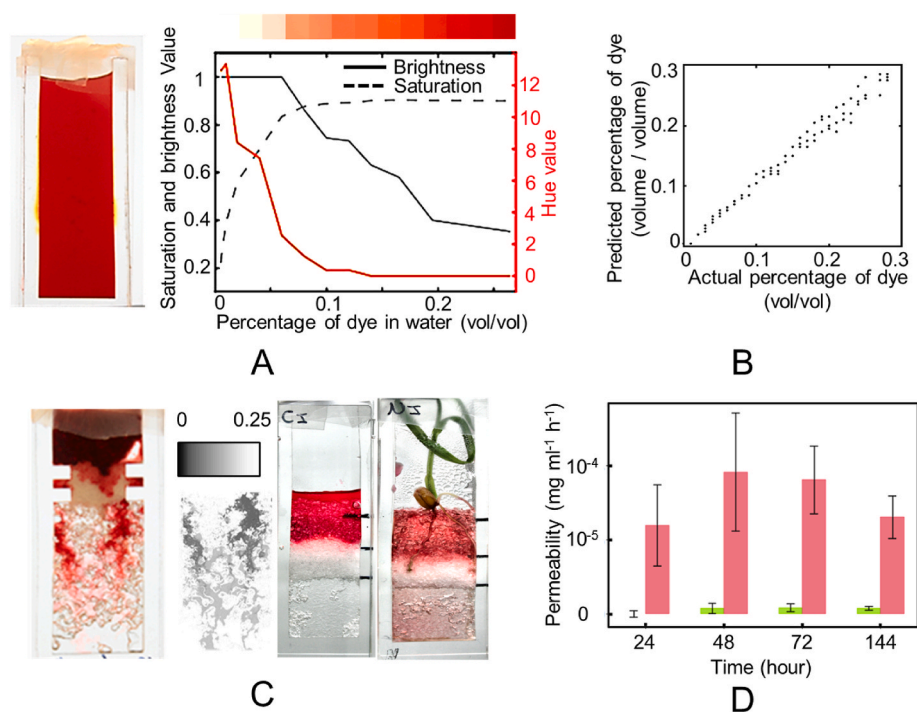


Fig. 5. Application of microcosm chambers to the characterisation of water transport through dry transparent soil layers. (A, left) Photo of the dye-water solution at 0.18 % (vol/vol). (A, right) Relationship of dye concentration and recorded HSV values from the photo. (B) Correlations between experimental and predicted dye concentration. (C) The dye solution infiltrated through the dry transparent soil layer (left) can then be determined through the permeability of the dry transparent soil layer (Eq. (4)). (D) Results show the root facilitated the transport of water through the dry transparent soil layer (data shown mean permeability $\pm$ standard error).

used to confirm that the effect of the root was statistically significant ($\chi^2 = 24.237$, $df = 1$, P -value = 8.5×10^{-7}). By contrast, the colouration of the bottom soil layer in control treatments was not different from the initial conditions and permeability was not significantly different from 0.

These results show that microcosm experiments can significantly enhance our knowledge of water fluxes in the rhizosphere. Hydrologists have long established that plants modify water movements in soil through preferential flow (Bouma, 1981; Flury et al., 1994), which limits runoff and its associated soil erosion. However, different types of plants can have varying effects on water infiltration (Durán Zuazo & Rodríguez Pleguezuelo, 2008; Jačka et al., 2021), and the role of biological activity (roots and microorganisms secretion, microbial growth) in these processes is yet to be clarified (Naveed et al., 2019; Or et al., 2007).

5. Conclusion

This study has explored ways to refine and streamline the preparation of soil microcosm systems designed for live, *in situ* imaging of root soil interactions. The approach involves the rapid prototyping and reproduction of chamber designs in a laboratory setting, employing techniques such as 3D printing, laser cutting, and cast moulding of parts. Intensive testing of the system has revealed remarkable tunability, especially with regard to environmental conditions and the control of water content, enhancing the reproducibility and reliability of experiments. Furthermore, the exploration of various materials has identified several as being suitable for fabricating transparent soil substrates. Specific experiments introduced a heterogeneous distribution of soil water content and employed tracer dyes to quantify water infiltration. This study demonstrates that microcosm system can be applied to effectively address current knowledge gaps in understanding plant-soil interactions.

CRediT authorship contribution statement

Yangminghao Liu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Daniel Patko:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Alberto Lora de la Mata:** Methodology, Investigation. **Xingshui Dong:** Methodology, Investigation. **Emma Gomez Peral:** Methodology, Investigation, Formal analysis. **Xinhua He:** Writing – original draft, Supervision. **Bruno Ameduri:** Writing – review & editing, Supervision, Resources, Conceptualization. **Vincent Ladmiral:** Supervision, Resources, Conceptualization. **Michael P. MacDonald:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Lionel X. Dupuy:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Statement on the use of generative AI and AI assisted technologies in the writing process

No generative AI or AI-assisted technologies were used during the preparation of this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was funded by the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant number 647857-SENSOILS) as well as Biotechnology and Biological Sciences Research Council under Des-BL project (grant

number BB/T010657/1). We also acknowledge the funding from the Spanish Ministry of Science and Innovation (MICINN) under de project MICROCROWD (grant number PID 2020-112950RR-I00). We thank Mr Masahiro Ohkura from AGC Inc for the supply of Forblue™ free samples.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biosystemseng.2025.02.006>.

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- number BB/T010657/1). We also acknowledge the funding from the Spanish Ministry of Science and Innovation (MICINN) under de project MICROCROWD (grant number PID 2020-112950RR-I00). We thank Mr Masahiro Ohkura from AGC Inc for the supply of Forblue™ free samples.
- ## Appendix A. Supplementary data
- Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biosystemseng.2025.02.006>.
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